HRMS

Human Resource Management System

Solution Design Document

Metadata-Driven Architecture | Custom ERP | Built from Scratch

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1. Executive Summary

This document describes the complete Solution Design for a custom-built Human Resource Management System (HRMS), developed as a standalone custom ERP application built from scratch.

The core philosophy of this HRMS is to move from a time-based evaluation model to a value-based model — measuring employees by their professional capability, work quality, efficiency, asset ownership, and long-term contribution to the organization.

Version 2.0 of this document introduces the foundational architectural principle that governs the entire system:

> **CORE PRINCIPLE: Every business rule, scoring weight, formula, threshold, and policy in this system is metadata-driven. No business logic is hardcoded. All rules are stored in the database and managed through the Admin UI. Policy changes require zero code deployment.**

This document covers the project scope, metadata-driven architecture design, full configuration entity catalog, functional module specifications, and non-functional requirements.

2. Project Scope

2.1 In Scope

- Metadata-Driven Configuration Engine — the foundation layer all modules depend on
- Base Professional Value (BPV) Engine — configurable scoring engine for employee professional worth
- Outcome-Based Time Reporting — delivery-based reporting with configurable time blocks
- Dual-Component Salary Structure — BPV salary + time-reporting variable, configurable split
- Efficiency Evaluation and Scoring — configurable scoring and incentive tier mapping
- Employee Data Ownership Registry — full ownership metadata for all digital assets
- Documentation Compliance Management — configurable compliance rules per asset type
- Long-Term Reward and Royalty Management — configurable incentive engine
- Beneficiary Management — employee-designated beneficiaries for long-term rewards
- HR and Finance Administration Dashboard — all configuration, reporting, and oversight
- Reports and Analytics — productivity, efficiency, contribution, and reward analytics

2.2 Out of Scope

- Payroll disbursement and banking integrations (Phase 2)
- Recruitment and applicant tracking
- Leave and travel expense management
- Native mobile application (responsive web is in scope; mobile app is a future phase)
- Third-party HRMS integrations (Salesforce, SAP, Workday, etc.)

2.3 Assumptions

- All business rules, scoring criteria, thresholds, and policies are configured by HR/Finance admins through the Admin UI — no developer involvement needed for policy changes.
- All configuration changes are versioned with effective dates. Historical calculations use the configuration that was active at the time.
- Authentication is handled via role-based access control (RBAC) with SSO support.
- The system is cloud-hosted; provider to be confirmed.
- All monetary values and salary calculations are currency-agnostic — currency is a configuration setting.

3. Metadata-Driven Architecture

3.1 Core Design Principle

The entire HRMS is built on a metadata-driven architecture. This means that every business rule, scoring formula, policy threshold, allowed value, and calculation parameter is stored as configuration data in the database — not as hardcoded values in application code.

> Admin configures rules → System stores in configuration tables → Business logic reads config at runtime → Output is calculated dynamically

This architecture delivers the following guarantees:

- Zero code deployment required for policy changes — HR and Finance update rules through the Admin UI.
- Full version history of every configuration — changes are tracked with effective dates and the user who made the change.
- Historical accuracy — past calculations always use the configuration that was active at the time of calculation, even after rules change.
- New categories can be added at any time — new qualification types, skill categories, asset types, reward types — without touching code.
- Configurable per grade, department, designation, or organization unit — rules don't have to be uniform across the organization.

3.2 Configuration Versioning Model

Every configuration entity follows the same versioning pattern:

Field	Purpose
config_id	Unique identifier for the configuration record
effective_from	Date from which this configuration version is active
effective_to	Date until which this version is active (null = current)
is_active	Boolean flag for the current active version
created_by	Admin user who created this configuration
created_at	Timestamp of creation
change_reason	Mandatory reason log for every configuration change

When a configuration is updated, the old record's effective_to is set to today's date and a new record is inserted with effective_from = tomorrow. No configuration record is ever deleted.

4. Configuration Entity Catalog

The following tables define every configurable entity in the system. All are managed through the HR/Finance Admin Dashboard with full version control.

4.1 BPV Engine Configuration Entities

Config Entity / Table	Purpose	Who Configures
BPV Weight Config	Stores the percentage weight for each BPV scoring factor (education, experience, org profile, skills, certifications). Must sum to 100%. Versioned with effective dates.	HR Admin
Qualification Master	Defines every recognized qualification type and its base score contribution. HR can add new qualifications at any time.	HR Admin
Experience Band Master	Maps year ranges to score points (e.g., 0–2 yrs = 5 pts, 3–5 yrs = 10 pts). Bands and points are fully configurable.	HR Admin
Experience Type Master	Defines types of experience (domain, enterprise, global, leadership) and their individual weight contribution within the experience scoring factor.	HR Admin
Organization Type Master	Maps organization types (startup, Fortune 500, government, research, etc.) to score values. New org types can be added at any time.	HR Admin
Skill Category Master	Defines skill categories (programming languages, platforms, frameworks, tools, methodologies). Categories are configurable.	HR Admin
Skill Master	Stores individual skills under each category. New skills are added by HR or IT as the technology landscape evolves.	HR Admin / IT
Skill Proficiency Master	Defines proficiency levels (e.g., beginner, intermediate, expert) and the score multiplier for each level. Levels and multipliers are configurable.	HR Admin
Certification Master	Stores recognized certifications (AWS, Azure, PMP, CPA, etc.) with their score contribution. New certifications are added by HR.	HR Admin
BPV Score Cap Config	Defines the maximum allowable BPV score (optional ceiling). Can be set per designation or grade.	HR Admin

4.2 Time Reporting Configuration Entities

Config Entity / Table	Purpose	Who Configures
Time Block Master	Defines the allowed time reporting options	HR Admin

Config Entity / Table	Purpose	Who Configures
	(e.g., 1 hr, 2 hrs, 4 hrs, 6 hrs). HR can add or remove blocks without code changes.	
Task Category Master	Defines task categories with configurable benchmark durations. Benchmark times are used for efficiency scoring comparisons.	HR Admin / Manager
Quality Rating Master	Defines the quality rating scale (e.g., 1–5, or Poor / Acceptable / Good / Excellent) used by managers when reviewing task delivery.	HR Admin
Quality Criteria Master	Maps quality criteria checklist items to task categories. Different task types can have different quality review criteria.	HR Admin / Manager

4.3 Salary Structure Configuration Entities

Config Entity / Table	Purpose	Who Configures
Salary Grade Master	Defines employee grades and the Component 1 / Component 2 salary split ratio per grade (e.g., 85% BPV / 15% Time Reporting). Configurable per grade, department, or designation.	HR Admin / Finance
Salary Band Master	Maps BPV score ranges to salary ranges per grade. Defines the monetary value of each BPV score bracket. Currency is a configuration setting.	Finance Admin
Currency Config	Defines the active currency and formatting rules. Supports multi-currency for global organizations.	Finance Admin
Review Cycle Config	Defines how often BPV and salary components are reviewed and recalculated (monthly, quarterly, annually). Can differ by grade or department.	HR Admin

4.4 Efficiency Evaluation Configuration Entities

Config Entity / Table	Purpose	Who Configures
Efficiency Weight Config	Stores the weight of each efficiency input factor — task completion rate, quality score, and time efficiency. Weights sum to 100% and are fully configurable.	HR Admin
Efficiency Tier Master	Maps efficiency score ranges to incentive eligibility percentages (e.g., 90–100 = 25% bonus, 75–89 = 15%, below 75 = 0%). Tiers and percentages are configurable.	HR Admin / Finance
Efficiency Period Config	Defines the measurement period for efficiency scoring — monthly, quarterly, or annual. Configurable per department or designation.	HR Admin

4.5 Data Ownership Configuration Entities

Config Entity / Table	Purpose	Who Configures
Asset Type Master	Defines all recognized digital asset categories (source code, documents, dashboards, AI prompts, architecture docs, etc.). New types are added by HR/IT as needed.	HR Admin / IT
Asset Status Master	Defines valid lifecycle statuses for assets (Active, Under Review, Archived, Deprecated). Configurable.	HR Admin
Ownership Role Master	Defines ownership roles that can be assigned to an asset (Creator, Contributor, Reviewer, Approver, Owner). Roles and their responsibilities are configurable.	HR Admin

4.6 Documentation Compliance Configuration Entities

Config Entity / Table	Purpose	Who Configures
Documentation Type Master	Defines recognized documentation types (Implementation Guide, User Guide, Maintenance Instructions, Dependencies, Version History, etc.). New types can be added.	HR Admin / IT
Doc Requirement Config	Maps required documentation types to each asset category. Different asset types can have different required documentation checklists.	HR Admin / IT
Compliance Threshold Config	Defines the minimum documentation compliance score (%) before an asset is flagged as non-compliant. Configurable per asset type or department.	HR Admin
Non-Compliance Action Config	Defines the consequence of non-compliance — flag only, block deployment, affect efficiency score, notify manager. Actions are configurable.	HR Admin

4.7 Reward and Royalty Configuration Entities

Config Entity / Table	Purpose	Who Configures
Reward Type Master	Defines available reward types — annual bonus, innovation incentive, profit sharing, royalty, long-term recognition. New reward types can be added.	HR Admin / Finance
Reward Basis Master	Defines which value metrics can be used to justify a reward — revenue generated, cost	Finance Admin

Config Entity / Table	Purpose	Who Configures
	savings, productivity improvement, IP value, strategic impact, customer adoption.	
Reward Formula Config	Stores the calculation formula and parameters for each reward type. Fully configurable by Finance without code changes.	Finance Admin
Reward Period Config	Defines when rewards are assessed and paid — annual, quarterly, milestone-based, or on-demand. Configurable per reward type.	Finance Admin
Reward Contribution Config	Defines how rewards are split among Creator and Contributors when multiple employees own an asset. Split method is configurable (equal, by contribution %, by manager discretion).	Finance Admin
Reward Cap Config	Sets maximum reward amounts per employee per period. Optional ceiling configurable per grade or reward type.	Finance Admin

4.8 Beneficiary Configuration Entities

Config Entity / Table	Purpose	Who Configures
Beneficiary Type Master	Defines all valid beneficiary categories (spouse, children, legal heirs, NGOs, trusts, educational institutions, etc.). Configurable per organizational policy and jurisdiction.	HR Admin
Beneficiary Doc Config	Defines what proof documents are required per beneficiary type (ID proof, relationship proof, legal declaration, etc.). Configurable.	HR Admin / Legal
Jurisdiction Config	Defines applicable legal jurisdictions for reward payouts. Supports multi-jurisdiction for global organizations.	HR Admin / Legal

4.9 System-Level Configuration Entities

Config Entity / Table	Purpose	Who Configures
Role Master	Defines system roles and their display names (Employee, Manager, HR Admin, Finance Admin, System Admin).	System Admin
Permission Master	Defines granular permissions per module and action (view, create, edit, approve, configure). Configurable per role.	System Admin
Department Master	Organization department definitions. Used for configuration scoping.	HR Admin
Designation Master	Job designation definitions. Used for salary band and configuration scoping.	HR Admin
Notification Config	Configures what events trigger notifications (low compliance, reward payout, BPV	System Admin

Config Entity / Table	Purpose	Who Configures
	recalculation, etc.) and to whom.	
Audit Log Config	Defines which actions are logged in the audit trail and their retention period.	System Admin

5. System Overview

The HRMS is a web-based custom ERP application with a modular, metadata-driven architecture. The Configuration Engine sits at the foundation — all functional modules read their business rules from configuration at runtime.

ARCHITECTURAL FLOW: Admin Dashboard → Configuration Engine → Functional Modules → Analytics & Reports

5.1 Module Map

Module	Depends On Config Entities
Employee Master	Department Master, Designation Master, Skill Master, Qualification Master
BPV Engine	BPV Weight Config, all scoring Masters, Review Cycle Config
Qualification & Experience Scoring	Qualification Master, Experience Band Master, Experience Type Master, Org Type Master
Technical Skill & Certification Mgmt	Skill Category Master, Skill Master, Skill Proficiency Master, Certification Master
Outcome-Based Task Reporting	Task Category Master, Quality Rating Master, Quality Criteria Master, Time Block Master
Time Reporting	Time Block Master, Task Category Master
Dual-Component Salary Management	Salary Grade Master, Salary Band Master, Currency Config, Review Cycle Config
Productivity & Efficiency Analytics	Efficiency Weight Config, Efficiency Tier Master, Efficiency Period Config
Data Ownership Registry	Asset Type Master, Asset Status Master, Ownership Role Master
Asset & Documentation Management	Documentation Type Master, Doc Requirement Config, Compliance Threshold Config, Non-Compliance Action Config
Contribution Tracking	Asset Type Master, Reward Contribution Config
Reward & Incentive Management	Reward Type Master, Reward Basis Master, Reward Formula Config, Reward Period Config, Reward Cap Config
Beneficiary Management	Beneficiary Type Master, Beneficiary Doc Config, Jurisdiction Config
HR & Finance Admin Dashboard	All configuration entities — this is where they are managed
Reports & Analytics	All functional modules — cross-module reporting

6. Functional Module Specifications

6.1 Employee Master

The Employee Master is the central profile record for each employee. All other modules reference this record. All reference data fields (department, designation, qualification type, skill, certification) resolve against Master configuration entities.

Key Data Fields

- Personal: name, date of birth, contact details, emergency contact
- Employment: employee ID, designation, department, reporting manager, date of joining, status
- Professional profile: qualifications (from Qualification Master), experience entries, organization history (from Org Type Master)
- Skill profile: skills and proficiency levels (from Skill Master + Skill Proficiency Master), certifications (from Certification Master)
- BPV Score: computed by the BPV Engine, updated per Review Cycle Config
- Salary: Component 1 (BPV-based), Component 2 (time-reporting), split per Salary Grade Master
- Status: Active / On Notice / Exited — ownership records are retained permanently regardless of status

6.2 Base Professional Value (BPV) Engine

The BPV Engine calculates each employee's professional worth score using configuration entities loaded at runtime. No formula is hardcoded.

6.2.1 Runtime Formula

> **BPV Score = (Education Score × Edu Weight) + (Experience Score × Exp Weight) + (Org Score × Org Weight) + (Skill Score × Skill Weight) + (Certification Score × Cert Weight)**

All weights are read from BPV Weight Config at the time of calculation. Weights are versioned — historical recalculations use the weight that was active at the relevant date.

6.2.2 Scoring Inputs — All Metadata-Driven

- Education Score: sum of scores for each qualification held by the employee, where scores are read from Qualification Master.
- Experience Score: score is computed per experience type (domain, enterprise, global, leadership) using Experience Band Master and Experience Type Master weights.
- Organization Score: sum of scores for each previous organization, read from Organization Type Master.
- Skill Score: sum across all skills, calculated as Skill Base Score × Proficiency Multiplier, both read from Skill Master and Skill Proficiency Master.
- Certification Score: sum of scores for held certifications, read from Certification Master.
- Optional score cap applied if BPV Score Cap Config is set for the employee's grade.

6.2.3 Recalculation Rules

- BPV is recalculated automatically when an employee's qualification, experience, or skill profile is updated.
- Scheduled recalculation runs per Review Cycle Config (monthly, quarterly, or annually).
- Manual recalculation can be triggered by HR Admin with a mandatory reason log.
- Full BPV history is retained — every score value at every point in time is stored.
- Salary Component 1 is updated per Salary Band Master at each review cycle.

6.3 Outcome-Based Task and Time Reporting

This module replaces the traditional attendance-and-timesheet model. The focus is on what was delivered, not how many hours were occupied. All configurable parameters are runtime-loaded.

6.3.1 Task Assignment and Reporting

- Managers assign tasks with deliverables, quality criteria (from Quality Criteria Master), and expected completion date.
- Employees report: completion status, quality self-assessment (from Quality Rating Master), any blockers, and time consumed.
- Managers review delivery and rate quality using the Quality Rating Master scale.
- Accepted, rejected, or returned-for-revision statuses are supported.

6.3.2 Time Blocks

Available time reporting options are loaded from Time Block Master at runtime. Default configuration: 1 hr, 2 hrs, 4 hrs, 6 hrs. HR Admin can modify this list without code changes.

6.3.3 Benchmark Time

Each task is assigned a category (from Task Category Master). Each category has a configurable benchmark duration. This benchmark is used by the Efficiency Engine to compare actual time reported against expected time.

6.4 Dual-Component Salary Structure

The salary structure is entirely driven by configuration. No salary amount or split ratio is hardcoded.

Component 1 — BPV Salary

- Percentage of total compensation — read from Salary Grade Master for the employee's grade.
- Salary amount derived from BPV Score using Salary Band Master (score range → salary range mapping).

- Updated at each review cycle per Review Cycle Config.

Component 2 — Time Reporting Value

- Remaining percentage of total compensation — defined as 100% minus Component 1 split.
- Computed based on accepted time entries for the reporting period.
- Rate per time block and computation method configurable in Salary Grade Master.

6.5 Efficiency Evaluation

The Efficiency Engine scores employee performance using three configurable input factors. All weights and tier mappings are loaded from configuration at runtime.

6.5.1 Efficiency Score Formula

> **Efficiency Score = (Completion Rate × CR Weight) + (Quality Score × QS Weight) + (Time Efficiency × TE Weight)**

All three weights are read from Efficiency Weight Config. They must sum to 100% and can be adjusted by HR Admin at any time.

6.5.2 Incentive Tier Mapping

Efficiency Score is mapped to incentive eligibility percentage using Efficiency Tier Master. Example configuration (fully editable):

Score Range	Incentive Eligibility	Notes
90 – 100	25% of annual salary	Top performer tier
75 – 89	15% of annual salary	High performer tier
60 – 74	5% of annual salary	Standard performer
Below 60	0%	Review required

6.6 Employee Data Ownership Registry

Every digital asset created by an employee is registered with full ownership metadata. Asset types are driven by Asset Type Master — the system supports any asset category HR/IT defines.

6.6.1 Asset Registration

- Creator registers the asset — selects asset type (from Asset Type Master), enters ownership details.
- Ownership roles assigned per Ownership Role Master (Creator, Contributor, Reviewer, Approver, Owner).
- Version history is maintained automatically on every asset update.

6.6.2 Post-Departure Rule

- On employee exit, system flags all assets owned by the employee for handover review.
- Operational ownership transfers to the organization.
- Historical authorship and accountability remain permanently attributed to the individual — immutable.

6.7 Documentation Compliance Management

Every registered asset must satisfy the documentation checklist defined in Doc Requirement Config for its asset type. Compliance is automatically calculated and monitored.

6.7.1 Compliance Calculation

- System checks each required documentation item (from Doc Requirement Config) against what has been submitted for the asset.
- Compliance Score = (Submitted Items / Required Items) × 100.
- If score falls below the threshold defined in Compliance Threshold Config, asset is flagged as non-compliant.
- Consequence of non-compliance is determined by Non-Compliance Action Config — flag, notify manager, block, or affect efficiency score.

6.8 Long-Term Reward and Royalty Management

The Reward Engine calculates and manages long-term incentives tied to the business value of employee-created assets. All formulas, reward types, and payment schedules are configuration-driven.

6.8.1 Reward Calculation Flow

- HR/Finance nominates an asset for reward based on a value metric (from Reward Basis Master).
- The system applies the formula from Reward Formula Config for the selected reward type.
- Reward amount is split across Creator and Contributors per Reward Contribution Config.
- Reward Cap Config is applied if a ceiling is configured for the grade or reward type.
- Payment is scheduled per Reward Period Config.

6.9 Beneficiary Management

Employees nominate beneficiaries for eligible long-term rewards. All beneficiary types and documentation requirements are metadata-driven.

- Beneficiary types resolved from Beneficiary Type Master — HR configures which types are allowed.
- Required documents per type defined in Beneficiary Doc Config.
- Jurisdiction rules per Jurisdiction Config — supports multi-jurisdiction organizations.

- Employees can update beneficiary nominations at any time (or within windows defined by HR policy config).
- Default rule: if no beneficiary is designated, reward routing is flagged to HR for manual resolution.

7. Non-Functional Requirements

7.1 Configuration Engine Performance

- Configuration data is cached in application memory at startup and refreshed on admin update — no per-request database reads for config.
- Cache invalidation is triggered automatically when any config entity is updated.
- BPV recalculation for a single employee must complete within 5 seconds.

7.2 Security

- Role-Based Access Control — configuration entities are accessible only to authorized admin roles.
- All configuration changes are logged in the audit trail: who, what, when, and the reason.
- No configuration record is ever deleted — only versioned and superseded.
- SSO support for enterprise authentication.

7.3 Scalability

- Configuration engine supports unlimited configuration entities — no code change needed for new entity types.
- System must support up to 500 concurrent users without degradation.
- Asset Ownership Registry must support up to 1 million asset records.

7.4 Data Integrity

- All ownership records, BPV histories, configuration versions, and reward calculations are append-only — immutable logs.
- Historical calculations always use the configuration version that was active at the time of calculation.
- No permanent deletion of any business record — archival only.

7.5 Auditability

- Every configuration change is logged with: changed field, old value, new value, effective date, changed by, reason.
- Every BPV recalculation is logged with: snapshot of config used, input values, output score, timestamp.
- Every reward calculation is logged with: formula version, inputs, output, approver.

8. User Roles and Access

Role	Configuration Access	Operational Access
Employee	None	Own profile, task reporting, time logging, asset registration, beneficiary nomination
Manager	Task Category benchmarks (own team)	Assign tasks, review delivery, rate quality, view team efficiency
HR Admin	All HR config entities — BPV, scoring, grades, tiers, asset types, compliance, beneficiary types	All employee profiles, reward nominations, compliance reports
Finance Admin	Salary band, reward formula, reward cap, currency, jurisdiction	Salary reports, reward calculations, beneficiary payout routing
System Admin	All configuration entities	User management, audit log, system-wide settings, notification config

9. Expected Business Benefits

- Rewards professional expertise and knowledge rather than time spent.
- Encourages faster, high-quality task completion through Efficiency Scoring.
- Aligns compensation with real professional value via the BPV Engine.
- Improves accountability for all organizational digital assets through the Ownership Registry.
- Ensures knowledge and authorship are never lost — even after an employee exits.
- Enforces documentation discipline, reducing knowledge debt and maintenance risk.
- Strengthens long-term employee engagement through configurable, asset-linked rewards.
- Policy changes are fully in HR/Finance control — no developer dependency for rule updates.
- Scales with organizational growth — new skill types, asset categories, reward types, and jurisdictions are added through configuration, not code.

10. Document Control

Document Title	HRMS — Solution Design Document
Version	v2.0
Previous Version	v1.0 — Initial SDD
Key Change in v2.0	Metadata-Driven Architecture added as foundational principle; full Configuration Entity Catalog added; all module specs updated to reference config entities
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